

The reproducibility package for “Forecasting Systemic Risk Measures Using a Dynamic Semiparametric Approach Based on the Asymmetric Laplace Distribution”

Date of Assembly

This reproducibility package was completed on **October 20, 2025**.

Data

In the empirical analysis, we use the CSI 300 Index and the 31 Shenwan primary industry indices, along with the total market values of each primary industry. All data are sourced from the CHOICE database.

Computing Environment

The provided MATLAB and R code replicates the results presented in the paper. The code was run using **MATLAB R2024b** and **R 4.5.1**.

- To run the MATLAB program, please first download **James LeSage's Econometrics Toolbox**, **Kevin Sheppard's GARCH Toolbox** and **Andrew Patton's "Handbook of Economic Forecasting" Toolbox for MATLAB** ([Andrew Patton's home page](#)) . In addition, the **Parallel Computing Toolbox** of MATLAB is also required.

- The following R packages are required:

tseries

MCS

superheat

igraph

iterators

parallel

foreach

doParallel

Experimental Environment

All experiments were conducted with the following specifications:

- **CPU:** Intel Xeon Silver 4216 2.10GHz (32 cores)
- **RAM:** 256 GB
- **GPU:** NVIDIA RTX™ A5000 GPU
- **Operating System:** Linux (Ubuntu 22.04)

Figures and Tables

The following two tables introduce how to generate the specific tables and figures presented in the paper. Each table and figure requires preprocessing steps. To fully reproduce all experimental results, the MATLAB code in the subfolders (**i2sys**, **sys2i**, **network**) of the **GASCoC_Matlab** folder should be executed sequentially, as outlined in Table 1. All output .mat files must be saved in the **Matlab_results** folders. Afterward, all figures and tables can be generated using the procedures described in Table 2.

Table 1 Roadmap for the Complete Reproduction of Experimental Results

Steps	i2sys	sys2i	network
1	A01_CoCAViaRi2sys_full.m	A01_CoCAVGASs2i_full	A01_CoESnetcalculate
2	A02_CopulaGaussiani2s_full	A02_CopulaGaussians2i_full	
3	A03_CopulaPOTi2s_full	A03_CopulaPOTs2i_full	
4	A04_DCCGaussiani2s_full	A04_DCCGaussians2i_full	
5	A05_DCCPOTi2s_full	A05_DCCPOTs2i_full	
6	A06_CoCAViaRi2sys_OOS	A06_CoCAViaRsys2i_OOS	
7	A07_CopulaGausi2s_OOS	A07_CopulaGaus2i_OOS	
8	A08_CopulaPOTi2s_OOS	A08_CopulaPOTs2i_OOS	
9	A09_DCCGausi2s_OOS	A09_DCCGauss2i_OOS	
10	A10_DCCPOTi2s_OOS	A10_DCCPOTs2i_OOS	
11		A11_portfolioAL	
12		A12_portfolioALconst	
13		A13_portfolioGARCH	

Table 2 Reproducibility Roadmap Based on Experimental Results
(in the order in which the table and figure appear in the paper)

Table/Figure	Software: Folder: Code
Table 1	Matlab: Matlab_results: A01_Table1.m
Table 2	Matlab: Matlab_results: A02_Table2.m
Table 3	① Matlab: Matlab_results: A03_Table3_1.m ② R: CoVaRCoes: A01_Table3_2.R
Table 4	① Matlab: Matlab_results: A04_Table4andFigure1_1.m ② R: CoVaRCoes: A02_Table4_2.R
Figure 1	① Matlab: Matlab_results: A04_Table4andFigure1_1.m ② R: CoVaRCoes: A03_Figure1_2.R
Figure 2	Matlab: Matlab_results: A05_Figure2.m
Table 5	① Matlab: Matlab_results: A06_Table5andFigure3_1.m ② R: CoVaRCoes: A04_Table5_2.R
Figure 3	① Matlab: Matlab_results: A06_Table5andFigure3_1.m ② R: CoVaRCoes: A05_Figure3_2.R
Table 6	Matlab: Matlab_results: A07_Table6.m
Figure 4	Matlab: Matlab_results: A08_Figure4.m
Table 7	Matlab: Matlab_results: A09_Table7.m